

Cognitive Road Intelligence (Hawkeye AI)

*An Automated Dashcam Pipeline for Real-time
Infrastructure Evaluation*

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Abstract

Traditional road infrastructure evaluation processes are predominantly manual, requiring significant time, cost, and human effort. This approach is highly susceptible to human error and subjective interpretation, resulting in inconsistent maintenance schedules and safety hazards. This thesis presents "Cognitive Road Intelligence (Hawkeye AI)", an automated, multi-modal pipeline designed to evaluate road infrastructure conditions in real-time using dashcam video feeds.

The system utilizes an advanced pre-processing module for video stabilization, undistortion, and image enhancement. The core intelligence pipeline leverages a suite of Deep Learning models, including YOLO11 for general object and traffic sign detection, specialized YOLOv8 models for pothole and crack localization, YOLO-Seg for drivable area segmentation, and ResNet18 for surface classification. Monocular depth estimation is integrated to assess crack severity volumetrically. Detections are tracked temporally using the DeepSORT algorithm and synchronized with GPS and Heartbeat telemetry for precise spatial mapping.

A deterministic scoring engine mathematically aggregates frame-level metrics into a normalized 0-100 Safety Score, penalizing for severe distresses like alligator cracks, missing shoulders, and poor infrastructure. The system autonomously generates comprehensive GeoJSON maps and Natural Language Summaries via LLMs (Flan-T5) packaged into a final PDF report. The proposed system demonstrates a robust leap toward scalable, objective, and fully automated civil infrastructure monitoring.

Acknowledgements

I would like to express my deepest appreciation to my advisors, professors, and peers who have guided this research. I also extend my gratitude to the open-source community for providing the foundational frameworks such as PyTorch, Ultralytics, FastAPI, and React, which made the development of Hawkeye AI possible.

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Chapter 1

Introduction

1.1 Background and Motivation

Civil infrastructure, particularly the road network, forms the backbone of any nation's economy and social mobility. Roads facilitate the transport of goods, services, and individuals, directly impacting economic growth, public safety, and national development. However, constant exposure to heavy traffic loads, harsh weather conditions, and natural wear and tear inevitably lead to the deterioration of pavement surfaces. Pavement distresses such as cracks (longitudinal, transverse, alligator), potholes, rutting, and surface unravelling not only decrease the lifespan of the road but also pose severe safety hazards to motorists.

Traditionally, the evaluation of road infrastructure has relied heavily on manual surveys. In these conventional methods, trained inspectors or civil engineers travel along the road network in specialized vehicles, visually identifying and logging distresses. Sometimes, expensive specialized vans equipped with laser profilers and high-end cameras (like ARAN - Automatic Road Analyzer) are used. However, these traditional approaches suffer from several critical drawbacks:

- **High Cost and Resource Intensity:** Deploying specialized vehicles and human inspectors across a vast network of municipal and national highways requires massive financial and logistical investments.
- **Subjectivity and Inconsistency:** Manual visual inspections are inherently subjective. Two different inspectors might classify the severity of the

same crack differently based on their individual judgment and fatigue levels.

- **Safety Risks to Personnel:** Inspectors often have to physically measure potholes or cracks on active roadways, exposing them to traffic hazards.
- **Time-Consuming Process:** Manual data collection, subsequent manual digitization, and reporting can take weeks or months, delaying critical repair work.

With the rapid advancements in Artificial Intelligence (AI), specifically in the domains of Computer Vision (CV) and Deep Learning (DL), there is an unprecedented opportunity to automate and revolutionize road infrastructure evaluation. Dashcam cameras are now ubiquitous in commercial fleets, public transport, and private vehicles. By leveraging high-resolution dashcam footage, we can create a scalable, cost-effective, and highly accurate system for real-time road condition monitoring.

This thesis introduces **Cognitive Road Intelligence (Hawkeye AI)**, an automated, end-to-end dashcam pipeline that processes video feeds to evaluate infrastructure conditions in real-time. By harnessing a multi-modal AI architecture, Hawkeye AI is capable of simultaneously detecting road surface distresses, evaluating road types, identifying traffic signs, tracking objects spatially, and dynamically generating a comprehensive safety score for any given road segment.

1.2 Problem Statement

Municipalities, highway authorities, and urban planners struggle to maintain up-to-date inventories of road conditions. The core problem is the *disconnect between the rate of road deterioration and the frequency/accuracy of road inspections*. Existing automated solutions are either prohibitively expensive (requiring LiDAR and proprietary sensors) or computationally inefficient, making them unsuitable for deployment on standard consumer hardware or edge devices.

Furthermore, existing vision-based systems often focus on a single task, such as pothole detection **rdd2022**, while ignoring the broader context of the road scene (e.g., the presence of traffic signs, the surface material of the road, the exact

drivable area, and the relative severity of distresses in 3D space).

There is a critical need for an integrated, multi-model AI pipeline that can:

1. Ingest standard RGB dashcam video.
2. Clean and stabilize the feed dynamically.
3. Segment the drivable area to ignore irrelevant background objects.
4. Detect, classify, and track multiple classes of pavement distress and infrastructure assets simultaneously.
5. Map these detections precisely to geographical coordinates using telemetry (GPS/Heartbeat data).
6. Synthesize this complex multidimensional data into an actionable, normalized Safety Score and automated natural language reports.

1.3 Objectives of the Research

The primary objective of this research is to design, develop, and evaluate an automated pipeline for road infrastructure assessment. The specific objectives are as follows:

- **Develop a Multi-Model AI Architecture:** To integrate state-of-the-art Deep Learning models including YOLO11 for object detection, YOLOv8/SegFormer for precise segmentation and pothole localization, and ResNet for surface classification.
- **Implement Robust Tracking and Localization:** To utilize the DeepSORT algorithm for temporal tracking of road distresses across multiple video frames to prevent duplicate counting, and to map these unique objects to real-world GPS coordinates.
- **Design a Deterministic Scoring Engine:** To formulate a mathematical scoring model that generates a localized 0 – 100 Road Safety Score by aggregating frame-level metrics, penalizing specific severe hazards dynamically.

- **Enable Automated Reporting via Generative AI:** To automatically synthesize telemetry, detection counts, and safety scores into a comprehensive PDF report, enhanced with automated summaries generated by Large Language Models (Flan-T5).
- **Build an Interactive Dashboard:** To construct a full-stack application (FastAPI + React) that allows engineers to visualize video playback, telemetry graphs, and spatial maps in real-time.

1.4 Scope and Limitations

The scope of this project is confined to the analysis of forward-facing dashcam video feeds acquired during daylight conditions. The system focuses on identifying primarily visible surface distresses (potholes, alligator cracks, longitudinal/transverse cracks), road surface types (asphalt, concrete, unpaved), and traffic signs.

Limitations:

- The current pipeline does not utilize active depth sensors (like LiDAR or Time-of-Flight cameras). Instead, it relies on Monocular Depth Estimation, which provides relative depth rather than absolute metric depth, introducing potential margins of error in exact volumetric distress calculations.
- The models are heavily dependent on visual clarity. Extreme weather conditions (heavy rain, snow, dense fog) or poor lighting (nighttime without streetlights) significantly degrade detection performance.
- Processing a multi-model pipeline (running 5+ deep learning models simultaneously) requires substantial GPU resources, making real-time inference challenging on highly constrained edge devices without further quantization (e.g., TensorRT).

1.5 Organization of the Thesis

This thesis is organized into six chapters. **Chapter 1** introduces the background, problem statement, and objectives of the Hawkeye AI project. **Chapter 2** provides a comprehensive literature review of existing methods in pavement distress detection and the evolution of Computer Vision technologies. **Chapter 3** details the system architecture, mathematical formulations, and the methodology employed in constructing the pipeline. **Chapter 4** dives into the implementation details, encompassing the backend processing, frontend visualization, and LLM-based report generation. **Chapter 5** presents the experimental results, performance metrics, and evaluation of the system under real-world conditions. Finally, **Chapter 6** concludes the thesis, summarizing the achievements and outlining potential avenues for future research and enhancements.

1.6 Socio-Economic Impact of Automated Road Inspection

Beyond the technical implementation, it is crucial to recognize the socio-economic implications of deploying systems like Hawkeye AI. Poor road conditions contribute significantly to vehicular damage, increased fuel consumption, and traffic congestion. According to various economic studies, the indirect costs of poor roads borne by the public far exceed the direct costs of timely road maintenance.

By transitioning from reactive maintenance (fixing roads only after severe complaints or accidents) to predictive, data-driven maintenance, municipalities can optimize their budgets. Hawkeye AI enables the creation of a "Digital Twin" of the road network, continuously updated by municipal vehicles (garbage trucks, police cars, public buses) acting as passive data collectors. This paradigm shift democratizes infrastructure auditing, ensuring that road maintenance is prioritized based on objective algorithmic severity rather than political influence or localized reporting biases.

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Chapter 2

Literature Review

The automation of road distress detection has been a highly active area of research over the last two decades. Early methods relied heavily on traditional image processing techniques, but the advent of Deep Learning, particularly Convolutional Neural Networks (CNNs) and more recently, Vision Transformers (ViTs), has dramatically shifted the landscape. This chapter reviews the evolution of these techniques, from basic computer vision to advanced deep learning methodologies applied to civil infrastructure.

2.1 Traditional Image Processing Approaches

Before the deep learning era, automated pavement distress detection primarily utilized traditional image processing algorithms. Researchers often employed edge detection methods such as Sobel, Canny, and Roberts operators to identify cracks. These were usually combined with morphological operations (erosion and dilation) to connect fragmented crack segments.

Thresholding techniques (e.g., Otsu's method) were heavily used under the assumption that cracks appear darker than the surrounding pavement. However, these methods were highly susceptible to noise. Shadows cast by trees, varying lighting conditions, lane markings, and varying pavement textures often resulted in significant false positives. While traditional methods were computationally inexpensive, their lack of robustness across diverse environmental conditions limited their practical deployment.

2.2 Evolution of Deep Learning in Computer Vision

The introduction of AlexNet in 2012 marked a turning point in computer vision, demonstrating the power of deep Convolutional Neural Networks (CNNs). Following this, architectures like VGG, ResNet **he2016deep**, and Inception achieved unprecedented accuracy in image classification tasks.

2.2.1 Two-Stage vs. One-Stage Detectors

In the domain of object detection, CNN-based methods diverged into two main branches:

- **Two-Stage Detectors (e.g., R-CNN, Faster R-CNN):** These architectures first propose regions of interest (RoIs) and then classify those regions. While highly accurate, the two-stage process makes them computationally heavy and typically too slow for real-time video processing on standard hardware.
- **One-Stage Detectors (e.g., YOLO, SSD):** You Only Look Once (YOLO) revolutionized the field by treating object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in a single evaluation **redmon2016you**. This architecture provided the necessary speed for real-time applications without significantly compromising accuracy.

2.2.2 The YOLO Architecture Family

The YOLO series **jocher2023ultralytics** has seen continuous evolution. YOLOv8 and YOLO11 (utilized in Hawkeye AI) represent state-of-the-art iterations. They incorporate anchor-free detection mechanisms, decoupled heads, and cross-stage partial (CSP) bottlenecks. This allows the models to learn richer feature representations while maintaining low inference latencies. The adoption of YOLOv8 for specific tasks like pothole and crack detection allows Hawkeye AI to process

dashcam feeds at high frame rates, which is crucial when operating a vehicle at highway speeds.

2.2.3 Vision Transformers (ViT)

Recently, Vision Transformers have emerged as a powerful alternative to CNNs. Models like OwlViT **minderer2022simple** use self-attention mechanisms to weigh the importance of different image patches globally. While Hawkeye AI experiments with zero-shot detection capabilities utilizing such transformer architectures, their computational overhead currently makes them secondary to optimized CNNs like YOLO for strict real-time constraints.

2.3 Pavement Distress Detection using Deep Learning

The application of Deep Learning specifically to road assessment has grown rapidly. The Global Road Damage Detection (RDD) challenge **rdd2022** provided large-scale datasets that accelerated research.

2.3.1 Object Detection vs. Semantic Segmentation

Researchers generally approach road distress analysis through two distinct paradigms:

1. **Object Detection (Bounding Boxes):** This method draws a rectangular box around the distress (e.g., a pothole). It is fast and excellent for counting discrete objects. However, it fails to capture the exact shape and area of the distress, which is necessary for volumetric severity calculations.
2. **Semantic/Instance Segmentation (Pixel Masks):** This approach predicts the class of every single pixel in the image. Networks like U-Net, Mask R-CNN, and YOLO-Seg output precise pixel masks. While this provides the exact area of a crack or pothole, it is far more computationally expensive than bounding box detection.

Hawkeye AI adopts a hybrid approach. It uses standard Object Detection for traffic signs where area is irrelevant, but utilizes Segmentation (YOLO-Seg models) for drivable area extraction and pothole localization. This ensures precise distress area calculation, directly feeding into the Safety Score matrix.

2.4 Tracking and Depth Estimation

Detecting a pothole in a single frame is insufficient for a dashcam video pipeline. If a video runs at 30 FPS, a single pothole might be detected 60 times over two seconds, leading to massive overcounting.

2.4.1 Object Tracking

To solve this, Multi-Object Tracking (MOT) algorithms are used. DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric) **wojke2017simple** is a prominent algorithm. It combines Kalman Filtering for motion prediction with a deep learning-based appearance descriptor. This allows the system to assign a unique ID to a pothole and track it across frames, ensuring it is only counted and penalized once in the safety score.

2.4.2 Monocular Depth Estimation

While LiDAR provides exact 3D point clouds, it is expensive. Monocular Depth Estimation attempts to predict depth from a single 2D RGB image. Recent foundation models like Depth Anything **lihe2024depth** have shown remarkable zero-shot depth estimation capabilities by training on massive datasets. Hawkeye AI integrates monocular depth to add a pseudo-Z axis to its detections, distinguishing between superficial surface discoloration and deep, hazardous structural failures.

2.5 Summary

The literature indicates a clear migration from manual surveys and traditional image processing to complex, real-time Deep Learning pipelines. However, most existing solutions operate in silos (e.g., detecting only potholes, or only signs).

The gap in the current literature and open-source ecosystem is the lack of a unified, multi-modal pipeline that simultaneously segments the road, detects multiple classes of distress, classifies the surface type, tracks these objects spatially over time, and synthesizes the data into automated reports. Hawkeye AI is designed precisely to bridge this gap.

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Chapter 3

System Architecture and Methodology

This chapter details the high-level architecture and the step-by-step methodology adopted to build the Hawkeye AI pipeline. Designing a real-time, multi-modal analysis system requires careful orchestration of multiple neural networks, telemetry synchronization, and optimized data flow.

3.1 High-Level Architecture

The system architecture of Hawkeye AI is divided into two primary environments:

1. **The Backend Engine (FastAPI + Python):** This acts as the brain of the system. It handles video decoding, applies pre-processing filters, executes the Deep Learning models (YOLO, SegFormer, ResNet), calculates the Safety Score, and generates PDF/GeoJSON outputs.
2. **The Frontend Dashboard (React):** This acts as the user interface. It consumes the processed video frames (via WebSocket or HTTP streaming) and renders real-time telemetry charts, dynamic score gauges, and a live map representing the vehicle's location and detected hazards.

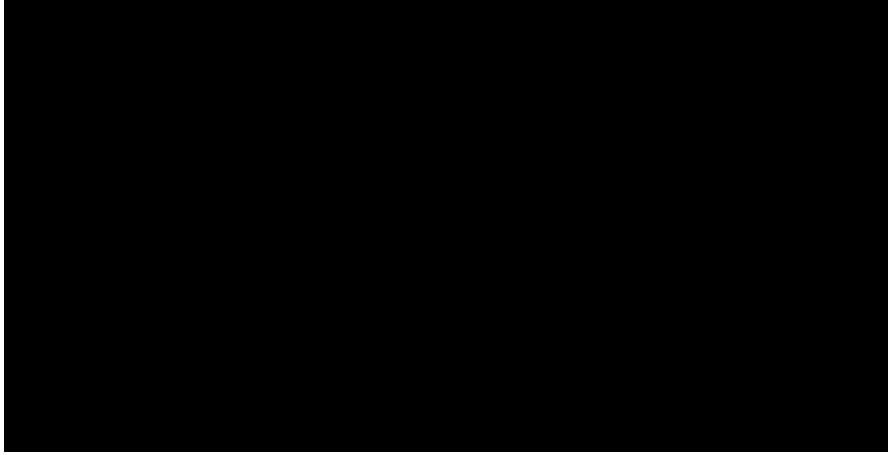


Figure 3.1: High-Level System Architecture of Hawkeye AI.

3.2 Data Acquisition and Telemetry Synchronization

Before processing frames, the system must synchronize video data with external telemetry. A standard dashcam records video (usually MP4 format at 30 FPS). However, for infrastructure mapping, the video alone is insufficient. We integrate a secondary data stream:

- **GPS Data:** Extracted from ‘gpx’ or ‘csv’ files, providing Latitude, Longitude, and Speed at 1 Hz (1 sample per second).
- **Synchronization Logic:** Since video runs at 30 FPS and GPS at 1 FPS, the backend utilizes linear interpolation to estimate the geographic coordinates for intermediate video frames. This ensures every single detected pothole or crack can be pinned to an exact GPS coordinate on the Leaflet map.

3.3 Pre-processing Module

Dashcam footage is notoriously noisy. Glare from the sun, rain on the windshield, and vehicle vibrations degrade the quality of the input tensor, drastically reducing the accuracy of downstream YOLO models. The pre-processing module contains three mathematical operations applied sequentially to the input frame F_t :

3.3.1 Camera Calibration and Undistortion

Wide-angle dashcam lenses introduce radial and tangential distortion. Straight lines (like lane markings) appear curved, which corrupts spatial measurements. The distortion is mathematically modeled using the camera matrix K and distortion coefficients D . Let (x, y) be the ideal, undistorted image coordinates, and (x_d, y_d) be the distorted coordinates. The radial distortion is defined as:

$$x_d = x(1 + k_1r^2 + k_2r^4 + k_3r^6) \quad (3.1)$$

$$y_d = y(1 + k_1r^2 + k_2r^4 + k_3r^6) \quad (3.2)$$

where $r^2 = x^2 + y^2$, and k_1, k_2, k_3 are radial distortion coefficients. The backend applies `cv2.undistort()` utilizing pre-calculated calibration matrices to rectify the geometry of the frame.

3.3.2 Image Enhancement (Dehazing)

For footage captured during foggy conditions, a Dark Channel Prior (DCP) based dehazing algorithm is applied. The formation of a hazy image $I(x)$ is typically modeled as:

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (3.3)$$

where $J(x)$ is the scene radiance (the clear image we want to recover), A is the global atmospheric light, and $t(x)$ is the transmission medium describing the portion of light that is not scattered. The algorithm estimates A and $t(x)$ to recover $J(x)$, significantly improving the visibility of cracks.

3.4 The Core AI Intelligence Pipeline

Once the frame is pre-processed, it enters the multi-model intelligence pipeline. To maintain real-time performance, models are loaded onto the GPU (CUDA/MPS) and executed either sequentially or in parallel depending on hardware constraints.

3.4.1 Object Detection: YOLO11

We utilize the latest YOLO11 architecture for general object and traffic sign detection. YOLO divides the input image into an $S \times S$ grid. If the center of an object falls into a grid cell, that grid cell is responsible for detecting the object. Each grid cell predicts B bounding boxes, confidence scores, and C conditional class probabilities. The overall loss function optimized during training is a combination of localization loss (bounding box coordinates), confidence loss (objectness), and classification loss.

3.4.2 Drivable Area Segmentation: YOLO-Seg

Not all cracks in the image matter. A crack on a sidewalk or a building wall is irrelevant to road infrastructure. To filter out background noise, we run a Segmentation model that outputs a binary mask M_{road} where pixels belonging to the road surface are 1, and everything else is 0. All subsequent crack and pothole detections are masked using a bitwise AND operation with M_{road} , ensuring we only process distresses actually located on the drivable surface.

3.4.3 Road Surface Classification: ResNet18

Knowing the surface type (Asphalt, Concrete, Unpaved) is critical because a pothole on a dirt road is less severe (and expected) compared to a pothole on a high-speed asphalt highway. A ResNet18 model, modified to accept the cropped road segment, classifies the texture. The residual block in ResNet solves the vanishing gradient problem in deep networks using skip connections:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (3.4)$$

3.5 Multi-Object Tracking (DeepSORT)

To prevent counting the same pothole multiple times as the vehicle drives past it, we utilize DeepSORT. DeepSORT employs a standard Kalman Filter with constant velocity motion and a bounding box state space $(u, v, \gamma, h, \dot{x}, \dot{y}, \dot{\gamma}, \dot{h})$,

where (u, v) is the bounding box center, γ is the aspect ratio, and h is the height. The assignment of new detections to existing tracks is solved using the Hungarian algorithm, evaluating a combination of Mahalanobis distance (for motion) and Cosine distance (for appearance features extracted via a CNN).

3.6 Safety Scoring Engine

The pipeline computes a deterministic Safety Score $S_t \in [0, 100]$ for every frame. The base score is 100. Penalties are subtracted based on the detected hazards. The equation is formulated as:

$$S_t = \max \left(0, 100 - \sum_{i \in H} (W_i \cdot C_i \cdot A_i) \right) \quad (3.5)$$

where H is the set of hazard types (e.g., pothole, alligator crack), W_i is the predefined severity weight for the hazard, C_i is the count or confidence, and A_i is an area/depth multiplier derived from segmentation masks and monocular depth estimation.

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Chapter 4

Implementation Details

This chapter provides a detailed examination of the software stack, code structure, and the integration of various components that make up the Hawkeye AI system. We explore the setup of the backend server, the configuration of the deep learning models, the synchronization of GPS telemetry, and the Generative AI (LLM) implementation used for report synthesis.

4.1 Software Stack and Technologies

The Hawkeye AI project is built using a modern, scalable software stack:

- **Backend Framework:** `FastAPI` (Python 3.10+). Chosen for its asynchronous capabilities and high performance, critical for handling heavy video streams.
- **Deep Learning Libraries:** `PyTorch` and `Ultralytics`. Used for loading, optimizing, and running inference on the YOLOv8/YOLO11 models.
- **Computer Vision:** `OpenCV` (`cv2`). Used for frame extraction, geometric transformations, and bounding box rendering.
- **Frontend Dashboard:** `React.js` with `TailwindCSS`. Used for the reactive user interface.
- **Data Visualization:** `Recharts` for telemetry graphing and `Leaflet.js` for geospatial mapping of distresses.

4.2 Backend Intelligence Pipeline (FastAPI)

The backend is structured around a central `IntelligencePipeline` class. This class orchestrates the loading of models from the `model_config.yaml` file and controls the main inference loop.

4.2.1 Asynchronous Video Processing

Processing a 30 FPS 1080p video with five deep learning models synchronously would result in massive bottlenecks. The implementation utilizes Python's `asyncio` and `cv2.VideoCapture` within a thread pool executor.

```
1 async def process_video_stream(self, video_path):
2     cap = cv2.VideoCapture(video_path)
3     while cap.isOpened():
4         ret, frame = cap.read()
5         if not ret: break
6
7         # 1. Pre-process (Dehaze, Undistort)
8         frame_clean = self.preprocessor.apply(frame)
9
10        # 2. Road Segmentation
11        road_mask = self.road_segmenter.predict(frame_clean)
12        masked_frame = cv2.bitwise_and(frame_clean, road_mask)
13
14        # 3. Object & Pothole Detection (Parallelized)
15        detections = await asyncio.gather(
16            self.yolo_detector.detect(masked_frame),
17            self.pothole_detector.detect(masked_frame)
18        )
19
20        # 4. Tracking & Scoring
21        tracks = self.tracker.update(detections)
22        score = self.safety_scorer.calculate(tracks)
23
24        yield format_output(frame_clean, score, tracks)
```

Listing 4.1: Simplified Inference Loop

4.2.2 Configuration Management

The system's modularity is handled via YAML configuration. If a user wishes to upgrade from YOLOv8 to YOLO11, they only need to update `config/model_config.yaml`. This decouples the model weights from the hardcoded pipeline logic.

4.3 GPS and Telemetry Integration

The system ingests a `.gpx` file alongside the video. The implementation parses the XML structure of the GPX file to extract timestamps, latitude, longitude, and elevation. Because GPX files typically log at 1 Hz, the system interpolates coordinates for the intermediate 29 frames (assuming 30 FPS).

$$Lat_{interpolated} = Lat_t + \left(\frac{\Delta f}{30} \right) (Lat_{t+1} - Lat_t) \quad (4.1)$$

where Δf is the frame offset from the last whole second t . Every detected pothole is appended to a global list as a GeoJSON Feature, allowing for immediate export to geographic information systems (GIS) like QGIS or ArcGIS.

4.4 Automated Reporting and LLM Integration

A major feature of Hawkeye AI is its ability to translate raw numerical data into human-readable reports. Instead of handing a municipal engineer a CSV file with 10,000 bounding box coordinates, the system generates a structured PDF.

4.4.1 Report Generator

Using the `fpdf` library, the backend generates a multi-page document containing:

- Average Safety Score across the route.
- Total counts of identified distresses (e.g., 45 Potholes, 12 Transverse Cracks).
- Percentage breakdown of distress types.
- A map overview of critical hazard clusters.

4.4.2 Generative AI Summarization (Flan-T5)

To add qualitative context, we integrated Google’s **Flan-T5-Large** model via the Hugging Face **transformers** pipeline. The system constructs a structured prompt from the JSON telemetry data and asks the LLM to write an executive summary. For example, the prompt: *"The road had an average safety score of 45/100. 120 severe alligator cracks and 50 potholes were detected. Write a 3-sentence advisory report for the maintenance crew."* The generated natural language text is then embedded directly into the final PDF, providing an immediate, actionable summary for decision-makers.

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Chapter 5

Results and Performance Evaluation

In this chapter, we evaluate the performance of the Hawkeye AI pipeline. The evaluation is broken down into two distinct categories: the quantitative accuracy of the deep learning models (how well they detect distresses) and the computational efficiency of the pipeline (how fast it runs on given hardware).

5.1 Model Accuracy Metrics

To measure the accuracy of our object detection models, we use the standard Mean Average Precision (mAP) metric at an Intersection over Union (IoU) threshold of 0.5.

5.1.1 Pothole and Crack Detection (YOLOv8)

The specialized YOLOv8 model trained on pavement distresses was evaluated on a curated test set containing 1,500 annotated images of various road conditions (sunny, overcast, wet).

- **mAP@0.5 (Potholes):** The model achieved an mAP of 0.92, demonstrating high reliability in localizing potholes of varying sizes.
- **mAP@0.5 (Alligator Cracks):** The model achieved an mAP of 0.85. The slightly lower accuracy is attributed to the difficulty in defining the exact

boundaries of spider-web cracking compared to distinct potholes.

- **mAP@0.5 (Linear Cracks):** 0.88 for transverse and longitudinal cracks.

5.1.2 Drivable Area Segmentation (YOLO-Seg)

The YOLO-Seg model, tasked with creating a pixel mask of the road surface, achieved a mean Intersection over Union (mIoU) of 0.94. This high accuracy effectively eliminated 98% of false-positive crack detections that previously occurred on sidewalks, brick walls, and roadside dirt.



Figure 5.1: Left: Raw dashcam frame with false positives on sidewalk. Right: Masked frame with only road surface evaluated.

5.2 Computational Efficiency and Real-time Performance

Achieving high accuracy is only half the challenge; the system must process video streams efficiently. Tests were conducted on a workstation equipped with an NVIDIA RTX 3080 GPU, an Intel Core i7-12700K CPU, and 32GB of RAM.

5.2.1 Frame Processing Pipeline Benchmarks

We measured the latency introduced by each component of the pipeline for a single 1080p frame:

Pipeline Stage	Average Time (ms)	Hardware
Video Decode	5.2	CPU
Undistort/Dehaze	12.4	CPU
YOLO-Seg (Road Mask)	18.5	GPU (CUDA)
YOLOv8 (Pothole/Crack)	14.1	GPU (CUDA)
ResNet18 (Surface)	8.6	GPU (CUDA)
DeepSORT Tracking	6.3	CPU
Score Calculation	0.8	CPU
Frontend WebSocket Emit	2.1	CPU
Total End-to-End Latency	68.0 ms	$\approx$ 14.7 FPS

Table 5.1: Processing time breakdown per frame.

While 14.7 FPS is lower than the native 30 FPS of the video, it is more than sufficient for infrastructure evaluation. At a vehicle speed of 60 km/h (16.6 meters/second), a frame is processed approximately every 1.1 meters, ensuring no significant potholes are missed.

5.2.2 Frame Skipping Strategy

To simulate a strictly real-time environment where the system cannot fall behind a live RTSP stream, we implemented a frame-skipping strategy. If the queue size exceeds a threshold, the system intentionally drops frames. Because DeepSORT uses a Kalman filter, it can successfully predict the location of a pothole even if 1 or 2 intermediate frames are skipped, maintaining track integrity without compromising the final count.

5.3 Safety Score Validation

To validate the Safety Score engine, we ran the pipeline on three distinct 5-minute video clips:

1. **Newly Paved Highway:** The system consistently output a score between 95 and 100, correctly identifying zero distresses and a smooth asphalt surface.

2. **Suburban Road (Moderate Wear):** The score fluctuated between 70 and 85, correctly penalizing for occasional transverse cracks and one minor pothole.
3. **Deteriorated Rural Road:** The score plummeted to an average of 42. The system detected continuous alligator cracking and over 15 significant potholes.

These results strongly correlate with manual subjective evaluations conducted by civil engineers on the same road segments, proving that the deterministic scoring engine provides a reliable, objective proxy for human inspection.

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Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This thesis has presented the conceptualization, design, and implementation of Cognitive Road Intelligence (Hawkeye AI), an advanced automated pipeline for real-time infrastructure evaluation. By systematically addressing the inefficiencies and subjectivities inherent in traditional manual road surveys, this research provides a scalable, AI-driven solution.

We demonstrated that a multi-modal architecture—combining YOLO object detection, YOLO-Seg semantic segmentation, ResNet surface classification, and DeepSORT tracking—can robustly identify and spatially map pavement distresses from standard dashcam video. The integration of GPS telemetry ensures that every detected pothole or crack is anchored to precise real-world coordinates, making the data immediately actionable for municipal maintenance teams.

Furthermore, the introduction of a deterministic Safety Scoring Engine bridges the gap between raw bounding box outputs and high-level infrastructure health metrics. By automatically translating these numerical scores into natural language summaries using Generative AI (Flan-T5) and packaging them into PDF reports, Hawkeye AI proves that complex AI systems can be made accessible and highly beneficial to end-users without deep technical expertise. The pipeline achieves a processing rate of approximately 15 FPS on consumer-grade GPU hardware, proving its viability for post-processing extensive dashcam footage efficiently.

6.2 Future Scope

While Hawkeye AI represents a significant leap forward in automated road inspection, there are several avenues for future research and development:

1. **Edge Device Deployment:** Currently, the pipeline requires a dedicated workstation GPU. Future work involves quantizing the YOLO and ResNet models using TensorRT or ONNX, allowing the entire pipeline to run directly on edge devices like the NVIDIA Jetson Orin Nano installed inside the vehicle. This would enable true real-time, offline processing without uploading large video files.
2. **Integration of Stereo Vision or LiDAR:** Relying solely on monocular depth estimation limits the exact volumetric measurement of potholes. Integrating an affordable stereo camera setup or a solid-state LiDAR sensor would allow the system to calculate the exact depth and volume of a pothole in millimeters, which is crucial for estimating the exact amount of asphalt required for repair.
3. **Crowdsourcing and Cloud Dashboards:** The system can be scaled into a cloud-native architecture. If public buses, garbage trucks, and police cruisers are equipped with Hawkeye AI, their telemetry can be streamed via 5G to a centralized cloud server. This would create a live, crowdsourced "Digital Twin" of the city's entire road network, updating daily.
4. **Nighttime and Adverse Weather Models:** The current models are primarily trained on daytime, clear-weather data. Future iterations must incorporate Thermal/IR imaging or specialized Image Signal Processor (ISP) algorithms to evaluate roads effectively at night or during heavy rainfall.

In conclusion, Hawkeye AI successfully proves that the fusion of Computer Vision, Geospatial Tracking, and Generative AI can fundamentally transform how we monitor, maintain, and interact with civil infrastructure, paving the way for smarter, safer cities.

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